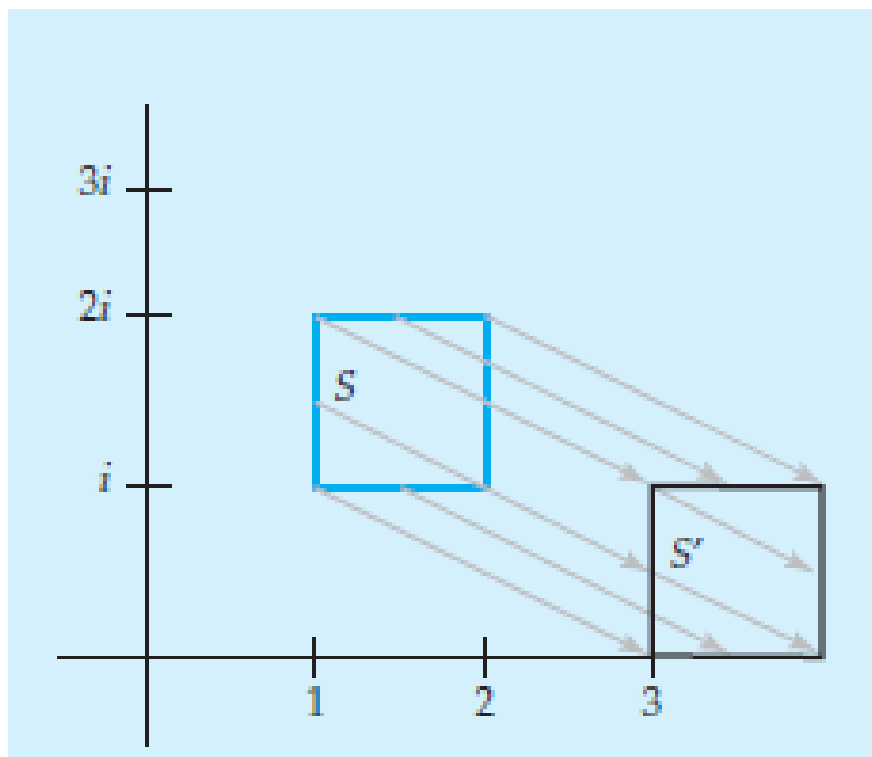




ANALYTIC FUNCTION

<ftp://10.220.20.26/academic/mce/maka>

Differentiability

Derivative: Suppose $z = x+iy$ and $z_0 = x_0+iy_0$; then the change in z_0 is the difference $\Delta z = z - z_0$ or $\Delta z = x - x_0 + i(y - y_0) = \Delta x + i\Delta y$. If a complex function $w = f(z)$ is defined at z and z_0 , then the corresponding change in the function is the difference $\Delta w = f(z_0 + \Delta z) - f(z_0)$. The **derivative** of the function f is defined in terms of a limit of the difference quotient $\Delta w / \Delta z$ as $\Delta z \rightarrow 0$.

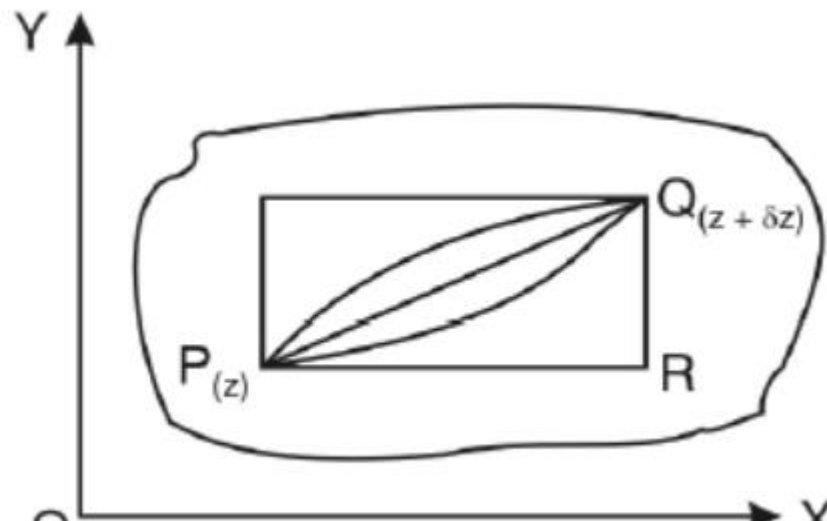
Differentiability

Let $f(z)$ be a single valued function of the variable z , then

$$f'(z) = \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}$$

provided that the limit exists and is independent of the path along which $\delta z \rightarrow 0$.

Let P be a fixed point and Q be a neighbouring point. The point Q may approach P along any straight line or curved path



Differentiability

~~Example 1:~~ Consider function $f z = 4x + y + i(4y - x)$ and discuss $\frac{df}{dz}$

Solution. Here, $f(z) = 4x + y + i(-x + 4y) = u + iv$
so $u = 4x + y$ and $v = -x + 4y$

$$\cancel{f(z + \delta z)} = 4(x + \delta x) + (y + \delta y) - i(x + \delta x) + 4i(y + \delta y)$$
$$\cancel{f(z + \delta z)} - f(z) = 4(x + \delta x) + (y + \delta y) - i(x + \delta x) + 4i(y + \delta y) - 4x - y + ix - 4iy$$
$$= 4\delta x + \delta y - i\delta x + 4i\delta y$$
$$\frac{\cancel{f(z + \delta z)} - f(z)}{\delta z} = \frac{4\delta x + \delta y - i\delta x + 4i\delta y}{\delta x + i\delta y}$$



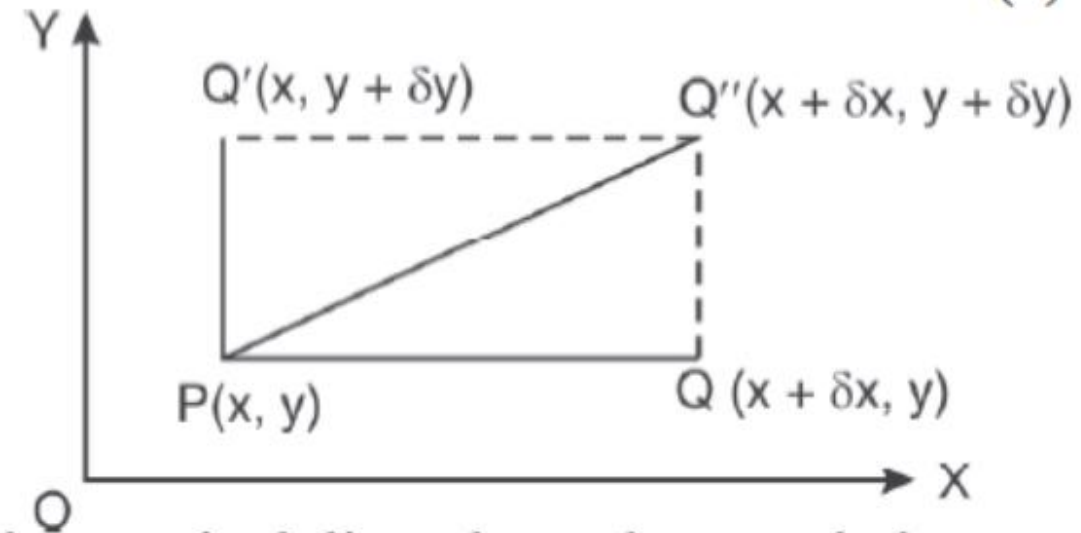
Differentiability

Example 1: Consider function $f z = 4x + y + i(4y - x)$ and discuss $\frac{df}{dz}$.

$$\Rightarrow \frac{\delta f}{\delta z} = \frac{4\delta x + \delta y - i\delta x + 4i\delta y}{\delta x + i\delta y} \quad \dots (1)$$

(a) Along real axis: If Q is taken on the horizontal line through $P(x, y)$ and Q then approaches P along this line, we shall have $\delta y = 0$ and $\delta z = \delta x$.

$$\frac{\delta f}{\delta z} = \frac{4\delta x - i\delta x}{\delta x} = 4 - i$$



Differentiability

Example 1: Consider function $f z = 4x + y + i(4y - x)$ and discuss $\frac{df}{dz}$

(b) **Along imaginary axis:** If Q is taken on the vertical line through P and then Q approaches P along this line, we have

$$z = x + iy = \underline{0 + iy}, \quad \underline{\delta z} = i\delta y, \quad \underline{\delta x = 0}.$$

Putting these values in (1), we have

$$\frac{\delta f}{\delta z} = \frac{\delta y + 4i\delta y}{i\delta y} = \frac{1}{i}(1 + 4i) = 4 - i$$

(c) Along a line $y = x$: If Q is taken on a line $y = x$.

$$z = x + iy = x + ix = (1 + i)x$$

$$\delta z = (1 + i)\delta x \quad \text{and} \quad \underline{\delta y = \delta x}$$

On putting these values in (1), we have

$$\frac{\delta f}{\delta z} = \frac{4\delta x + \delta x - i\delta x + 4i\delta x}{\delta x + i\delta x} = \frac{4 + 1 - i + 4i}{1 + i} = \frac{5 + 3i}{1 + i} = \frac{(5 + 3i)(1 - i)}{(1 + i)(1 - i)} = 4 - i$$

Differentiability

Example 1: Consider function $f z = 4x + y + i(4y - x)$ and discuss $\frac{df}{dz}$

In all the three different paths approaching Q from P , we get the same values of $\frac{\delta f}{\delta z} = 4 - i$.

In such a case, the function is said to be differentiable at the point z in the given region.

Differentiability

Example 2:

$$\text{If } f(z) = \begin{cases} \frac{x^3 y(y - ix)}{x^6 + y^2}, & z \neq 0, \\ 0, & z = 0 \end{cases} \text{ then discuss } \frac{df}{dz} \text{ at } z = 0.$$

Solution. If $z \rightarrow 0$ along radius vector $y = mx$

$$\begin{aligned} \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} &= \lim_{z \rightarrow 0} \left[\frac{\frac{x^3 y(y - ix)}{x^6 + y^2} - 0}{x + iy} \right] = \lim_{z \rightarrow 0} \left[\frac{-ix^3 y(x + iy)}{(x^6 + y^2)(x + iy)} \right] \\ &= \lim_{z \rightarrow 0} \left[\frac{-ix^3 y}{x^6 + y^2} \right] = \lim_{x \rightarrow 0} \left[\frac{-ix^3(mx)}{x^6 + m^2 x^2} \right] \\ &= \lim_{x \rightarrow 0} \left[\frac{-imx^2}{x^4 + m^2} \right] = 0 \end{aligned}$$

Differentiability

Example 2:

different values at $y=mx$ and $y=x^3$

But along $y = x^3$

$$\lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} = \lim_{z \rightarrow 0} \left[\frac{-ix^3 y}{x^6 + y^2} \right] = \lim_{x \rightarrow 0} \frac{-ix^3 (x^3)}{x^6 + (x^3)^2} = -\frac{i}{2}$$

In different paths we get different values of $\frac{df}{dz}$ i.e. 0 and $-\frac{i}{2}$. In such a case, the function is not differentiable at $z = 0$.

Differentiability

Example 3: Using the Rules of Differentiation

following formula

Differentiate:

$$(a) f(z) = 3z^4 - 5z^3 + 2z \quad (b) f(z) = \frac{z^2}{4z + 1} \quad (c) f(z) = (iz^2 + 3z)^5$$

Solution

~~(a)~~ Using the power rule (7), the sum rule (3), along with (2), we obtain

$$f'(z) = 3 \cdot 4z^3 - 5 \cdot 3z^2 + 2 \cdot 1 = 12z^3 - 15z^2 + 2.$$

~~(b)~~ From the quotient rule (5),

$$f'(z) = \frac{(4z + 1) \cdot 2z - z^2 \cdot 4}{(4z + 1)^2} = \frac{4z^2 + 2z}{(4z + 1)^2}.$$

~~(c)~~ In the power rule for functions (8) we identify $n = 5$, $g(z) = iz^2 + 3z$, and $g'(z) = 2iz + 3$, so that

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3).$$

Analytic at a Point

Definition: A complex function $w = f(z)$ is said to be **analytic at a point** z_0 if f is **differentiable** at z_0 and at **every point in some neighborhood of z_0** .

A function **f is analytic in a domain D** if it is analytic at every point in D . The phrase “**analytic on a domain D** ” is also used. Although we shall not use these terms.

✓ **Holomorphic:** a function f that is analytic throughout a domain D is called **holomorphic** or **regular**.

✓ **Entire Function:** A function that is analytic at every point z in the complex plane is said to be an **entire function**.

Analytic Function

Theorem 3.1 Polynomial and Rational Functions

- (i) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where n is a nonnegative integer, is an entire function.
- (ii) A rational function $f(z) = \frac{p(z)}{q(z)}$, where p and q are polynomial functions, is analytic in any domain D that contains no point z_0 for which $q(z_0) = 0$.

Singular Points: Since the rational function $f(z) = 4z / (z^2 - 2z + 2)$ is discontinuous at $1+i$ and $1-i$, f fails to be analytic at these points. Thus by (ii) of Theorem 3.1, f is not analytic in any domain containing one or both of these points. In general, a point z at which a complex function $w = f(z)$ fails to be analytic is called a **singular point** of f .

Analytic Function

An Alternative Definition of $f'(z)$ Sometimes it is convenient to define the derivative of a function f using an alternative form of the difference quotient $\Delta w/\Delta z$. Since $\Delta z = z - z_0$, then $z = z_0 + \Delta z$, and so (1) can be written as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}. \quad (12)$$

Theorem 3.2 Differentiability Implies Continuity

If f is differentiable at a point z_0 in a domain D , then f is continuous at z_0 .

Analytic Function

Theorem 3.3 L'Hôpital's Rule

Suppose f and g are functions that are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}. \quad (13)$$

Example 4: Using L'Hôpital's Rule

Solution If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2 + i) = 0$ and $g(2 + i) = 0$. The given limit has the indeterminate form $0/0$. Now since f and g are polynomial functions, both functions are necessarily analytic at $z_0 = 2 + i$. Using

Analytic Function

Example 4: Using L'Hôpital's Rule

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that (13) gives

$$\lim_{z \rightarrow 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8 + 12i} = \frac{3}{26} + \frac{1}{13}i.$$

Self Study: Using L'Hôpital's Rule

$$23. \quad \lim_{z \rightarrow i} \frac{z^7 + i}{z^{14} + 1}$$

$$24. \quad \lim_{z \rightarrow \sqrt{2} + \sqrt{2}i} \frac{z^4 + 16}{z^2 - 2\sqrt{2}z + 4}$$

$$25. \quad \lim_{z \rightarrow 1+i} \frac{z^5 + 4z}{z^2 - 2z + 2}$$

$$26. \quad \lim_{z \rightarrow \sqrt{2}i} z \frac{z^3 + 5z^2 + 2z + 10}{z^5 + 2z^3}$$

Analytic Function

Self Study: Determine the points at which the given function is not analytic.

$$27. f(z) = \frac{iz^2 - 2z}{3z + 1 - i}$$

$$28. f(z) = -5iz^2 + \frac{2+i}{z^2}$$

$$29. f(z) = (z^4 - 2iz^2 + z)^{10}$$

$$30. f(z) = \left(\frac{(4+2i)z}{(2-i)z^2 + 9i} \right)^3$$

Cauchy-Riemann Equations

Theorem 3.4 Cauchy-Riemann Equations

Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z = x + iy$. Then at z the first-order partial derivatives of u and v exist and satisfy the Cauchy-Riemann equations

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}} \quad \text{and} \quad \boxed{\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}. \quad (1)$$

Proof: Cauchy-Riemann Equations

Proof The derivative of f at z is given by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \quad (2)$$

By writing $f(z) = u(x, y) + iv(x, y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \quad (3)$$

Since the limit (2) is assumed to exist, Δz can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \rightarrow 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$\begin{aligned} f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y) + i[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}. \end{aligned} \quad (4)$$

Proof: Cauchy-Riemann Equations

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to x of u and v , respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point z , and that the derivative of f is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \quad (5)$$

We now let $\Delta z \rightarrow 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \quad (6)$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at z and that

$$f'(z) = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \quad (7)$$

Verifying : Cauchy-Riemann Equations

Example 5:

The polynomial function $f(z) = z^2 + z$ is analytic for all z and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point (x, y) in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

Criterion for Non-analyticity

If the Cauchy-Riemann equations are not satisfied at every point z in a domain D , then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in D .

Cauchy-Riemann Equations

THE NECESSARY CONDITION FOR $f(z)$ TO BE ANALYTIC

Theorem. *The necessary conditions for a function $f(z) = u + iv$ to be analytic at all the points in a region R are*

$$(i) \quad \underline{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}}$$

$$(ii) \quad \underline{\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \text{ provided } \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ exist.}$$

THE SUFFICIENT CONDITION FOR $f(z)$ TO BE ANALYTIC

Theorem. *The sufficient condition for a function $f(z) = u + iv$ to be analytic at all the points in a region R are*

$$(i) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$(ii) \quad \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ are } \underline{\text{continuous functions of } x \text{ and } y \text{ in region } R}.$$

Cauchy-Riemann Equations

- Remember:** 1. If a function is analytic in a domain D , then u, v satisfy $C - R$ conditions at all points in D .
2. $C - R$ conditions are necessary but not sufficient for analytic function.
3. $C - R$ conditions are sufficient if the partial derivative are continuous.

Example 6: Discuss the analyticity of the function $f(z) = z \bar{z}$.

Solution. $f(z) = z \bar{z} = (x + iy)(x - iy) = x^2 - i^2 y^2 = x^2 + y^2$

$$f(z) = x^2 + y^2 = u + iv.$$

$$u = x^2 + y^2, v = 0$$

At origin, $\frac{\partial u}{\partial x} = \lim_{h \rightarrow 0} \frac{u(0+h, 0) - u(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^2}{h} = 0$

$$\frac{\partial u}{\partial y} = \lim_{k \rightarrow 0} \frac{u(0, 0+k) - u(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{k^2}{k} = 0$$

Cauchy-Riemann Equations

Example 6: Discuss the analyticity of the function $f(z) = z \bar{z}$.

Also,

$$\frac{\partial v}{\partial x} = \lim_{h \rightarrow 0} \frac{v(0+h, 0) - v(0, 0)}{h} = 0$$
$$\frac{\partial v}{\partial y} = \lim_{k \rightarrow 0} \frac{v(0, 0+k) - v(0, 0)}{k} = 0$$

Thus,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Hence, C – R equations are satisfied at the origin.

$$f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} = \lim_{z \rightarrow 0} \frac{(x^2 + y^2) - 0}{x + iy}$$

Let $z \rightarrow 0$ along the line $y = mx$, then

$$f'(0) = \lim_{x \rightarrow 0} \frac{(x^2 + m^2 x^2)}{(x + imx)} = \lim_{x \rightarrow 0} \frac{(1 + m^2)x}{1 + im} = 0$$

Therefore, $f'(0)$ is unique. Hence the function $f(z)$ is analytic at $z = 0$.

Cauchy-Riemann Equations

Example 7: Determine whether $\frac{1}{z}$ is analytic or not?

Solution. Let $w = f(z) = u + iv = \frac{1}{z} \Rightarrow u + iv = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$

Equating real and imaginary parts, we get

$$u = \frac{x}{x^2 + y^2}, \quad v = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial u}{\partial x} = \frac{(x^2 + y^2) \cdot 1 - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2},$$

$$\frac{\partial u}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2}.$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2 + y^2)^2},$$

$$\frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Cauchy-Riemann Equations

Example 7: Determine whether $\frac{1}{z}$ is analytic or not?

Thus,
$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Thus C – R equations are satisfied. Also partial derivatives are continuous except at (0, 0).

Therefore $\frac{1}{z}$ is analytic everywhere except at $z = 0$.

✓✓ C-R EQUATION IN POLAR FORM

If $f(z) = u + iv$ is an analytic function where u and v are functions of r, θ and $z = re^{i\theta}$ then



$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

Example 8:

Show that $w = e^z$ is analytic everywhere in the complex plane. Hence, find $\frac{dw}{dz}$.

C-R EQUATION IN POLAR FORM

Example 8:

Show that $w = e^z$ is analytic everywhere in the complex plane. Hence, find $\frac{dw}{dz}$.

Solution

$$\begin{aligned}w &= e^z \\ \underline{u + iv} &= \underline{e^{x+iy}} \\ &= e^x e^{iy} \\ &= e^x (\cos y + i \sin y) \\ &= e^x \cos y + i e^x \sin y\end{aligned}$$

Comparing real and imaginary parts,

$$\begin{aligned}\underline{u} &= e^x \cos y, & \underline{v} &= e^x \sin y \\ \frac{\partial u}{\partial x} &= e^x \cos y, & \frac{\partial v}{\partial x} &= e^x \sin y \\ \frac{\partial u}{\partial y} &= -e^x \sin y, & \frac{\partial v}{\partial y} &= e^x \cos y \\ \therefore \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y}, & \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x}\end{aligned}$$

C-R EQUATION IN POLAR FORM

Example 8:

Show that $w = e^z$ is analytic everywhere in the complex plane. Hence, find $\frac{dw}{dz}$.



C-R equations are satisfied. Also e^x , $\cos y$ and $\sin y$ are continuous for all values of x and y .

Hence, e^z is analytic everywhere in the complex plane.

Since $w = u + iv$ is analytic everywhere,

$$\begin{aligned} \frac{dw}{dz} &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= e^x \cos y + i e^x \sin y \\ &= e^x (\cos y + i \sin y) \\ &= e^x e^{iy} \\ &= e^{x+iy} \\ &= e^z \end{aligned}$$

Harmonic Functions

Definition of Harmonic Function: A real-valued function ϕ of two real variables x and y that has continuous first and second-order partial derivatives in a domain D and satisfies Laplace's equation is said to be **harmonic** in D .

Theorem on Harmonic Functions: Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D . Then the functions $u(x, y)$ and $v(x, y)$ are harmonic in D . i. e.:

$$\left\{ \begin{array}{l} \frac{\partial^2 u}{\partial^2 x} + \frac{\partial^2 u}{\partial^2 y} = 0 \\ \frac{\partial^2 v}{\partial^2 x} + \frac{\partial^2 v}{\partial^2 y} = 0 \end{array} \right. \quad \text{or,} \quad \left\{ \begin{array}{l} \nabla^2 u = 0 \\ \nabla^2 v = 0 \end{array} \right.$$

Harmonic Functions

Definition of Harmonic Conjugate Functions : We have just shown that if a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its real and imaginary parts u and v are necessarily harmonic in D . Now suppose $u(x, y)$ is a given real function that is known to be harmonic in D . If it is possible to find another real harmonic function $v(x, y)$ so that u and v satisfy the Cauchy-Riemann equations throughout the domain D , then the function $v(x, y)$ is called a **harmonic conjugate** of $u(x, y)$. By combining the functions as $u(x, y) + iv(x, y)$ we obtain a function that is analytic in D .

Harmonic Functions

Example 9:

Example : Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic functions of (x, y) , but are not harmonic conjugates.

Solution. We have, $u = x^2 - y^2$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial^2 u}{\partial x^2} = 2, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial^2 u}{\partial y^2} = -2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0 \quad \text{laplace equation}$$

$u(x, y)$ satisfies Laplace equation, hence $u(x, y)$ is harmonic

$$v = \frac{y}{x^2 + y^2}, \quad \frac{\partial v}{\partial x} = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \frac{(x^2 + y^2)^2 (-2y) - (-2xy) 2(x^2 + y^2) 2x}{(x^2 + y^2)^4} \\ &= \frac{(x^2 + y^2)(-2y) - (-2xy) 4x}{(x^2 + y^2)^3} = \frac{6x^2 y - 2y^3}{(x^2 + y^2)^3} \end{aligned}$$

Example 9:

Harmonic Functions

$$= \frac{(x^2 + y^2)(-2y) - (-2xy)4x}{(x^2 + y^2)^3} = \frac{6x^2y - 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\partial v}{\partial y} = \frac{(x^2 + y^2) \cdot 1 - y(2y)}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \quad \dots (1)$$

$$\begin{aligned} \frac{\partial^2 v}{\partial y^2} &= \frac{(x^2 + y^2)^2(-2y) - (x^2 - y^2)2(x^2 + y^2)(2y)}{(x^2 + y^2)^4} = \frac{(x^2 + y^2)(-2y) - (x^2 - y^2)(4y)}{(x^2 + y^2)^3} \\ &= \frac{-2x^2y - 2y^3 - 4x^2y + 4y^3}{(x^2 + y^2)^3} = \frac{-6x^2y + 2y^3}{(x^2 + y^2)^3} \quad \dots (2) \end{aligned}$$

On adding (1) and (2), we get $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$

$v(x, y)$ also satisfies Laplace equations, hence $v(x, y)$ is also harmonic function.

But $\left[\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \right]$ and $\left[\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x} \right]$

Therefore u and v are not harmonic conjugates.

Proved.

✓✓✓ MLN THMSON METHOD

By this method $f(z)$ is directly constructed without finding v and the method is given below:
Since

$$z = x + iy \text{ and } \bar{z} = x - iy$$

∴

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

$$f(z) \equiv u(x, y) + iv(x, y) \quad \dots (1)$$

$$f(z) \equiv u\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) + iv\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right)$$

This relation can be regarded as a formal identity in two independent variables z and $\bar{z}$.
Replacing $\bar{z}$ by z , we get:

$$f(z) \equiv u(z, 0) + iv(z, 0)$$

Which can be obtained by replacing x by z and y by 0 in (1)

MLINE THMSON METHOD

1. check if it's harmonic
2. construct analytic function

Case I. When u is given

Step 1. Find $\frac{\partial u}{\partial x}$ and equate it to $\phi_1(x, y)$.

Step 2. Find $\frac{\partial u}{\partial y}$ and equate it to $\phi_2(x, y)$.

Step 3. Replace x by z and y by 0 in $\phi_1(x, y)$ to get $\phi_1(z, 0)$.

Step 4. Replace x by z and y by 0 in $\phi_2(x, y)$ to get $\phi_2(z, 0)$.

Step 5. Find $f(z)$ by the formula $f(z) = \int \{ \phi_1(z, 0) - i\phi_2(z, 0) \} dz + c$

MLINE THMSON METHOD

Case II. When v is given

Step 1. Find $\frac{\partial v}{\partial x}$ and equate it to $\psi_2(x, y)$.

Step 2. Find $\frac{\partial v}{\partial y}$ and equate it to $\psi_1(x, y)$.

Step 3. Replace x by z and y by 0 in $\psi_1(x, y)$ to get $\psi_1(z, 0)$.

Step 4. Replace x by z and y by 0 in $\psi_2(x, y)$ to get $\psi_2(z, 0)$.

Step 5. Find $f(z)$ by the formula

$$f(z) = \int \{ \psi_1(z, 0) + i\psi_2(z, 0) \} dz + c$$

MLINE THMSON METHOD

Example 10:

Show that $e^x (x \cos y - y \sin y)$ is a harmonic function. Find the analytic function for which $e^x (x \cos y - y \sin y)$ is imaginary part.

Solution. Here $v = e^x (x \cos y - y \sin y)$

Differentiating partially w.r.t. x and y , we have

$$\frac{\partial v}{\partial x} = e^x (x \cos y - y \sin y) + e^x \cos y = \psi_2(x, y), \quad (\text{say}) \quad \dots (1)$$

$$\frac{\partial v}{\partial y} = e^x (-x \sin y - y \cos y - \sin y) = \psi_1(x, y) \quad (\text{say}) \quad \dots (2)$$

double integral

$$\frac{\partial^2 v}{\partial x^2} = e^x (x \cos y - y \sin y) + e^x \cos y + e^x \cos y$$
$$= e^x (x \cos y - y \sin y + 2 \cos y) \quad \dots (3)$$

MILNE THOMSON METHOD

Example 10:

and

$$\frac{\partial^2 v}{\partial y^2} = e^x (-x \cos y + y \sin y - 2 \cos y) \quad \dots (4)$$

Adding equations (3) and (4), we have

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \Rightarrow v \text{ is a harmonic function.}$$

Now putting $x = z, y = 0$ in (1) and (2), we get

$$\psi_2(z, 0) = ze^z + e^z \quad \psi_1(z, 0) = 0$$

Hence by Milne-Thomson method, we have

$$\begin{aligned} f(z) &= \int [\psi_1(z, 0) + i\psi_2(z, 0)] dz + C \\ &= \int [0 + i(ze^z + e^z)] dz + C = i(ze^z - e^z + e^z) + C = iz e^z + C. \end{aligned}$$

This is the required analytic function.

Ans.

Thanks a lot ...